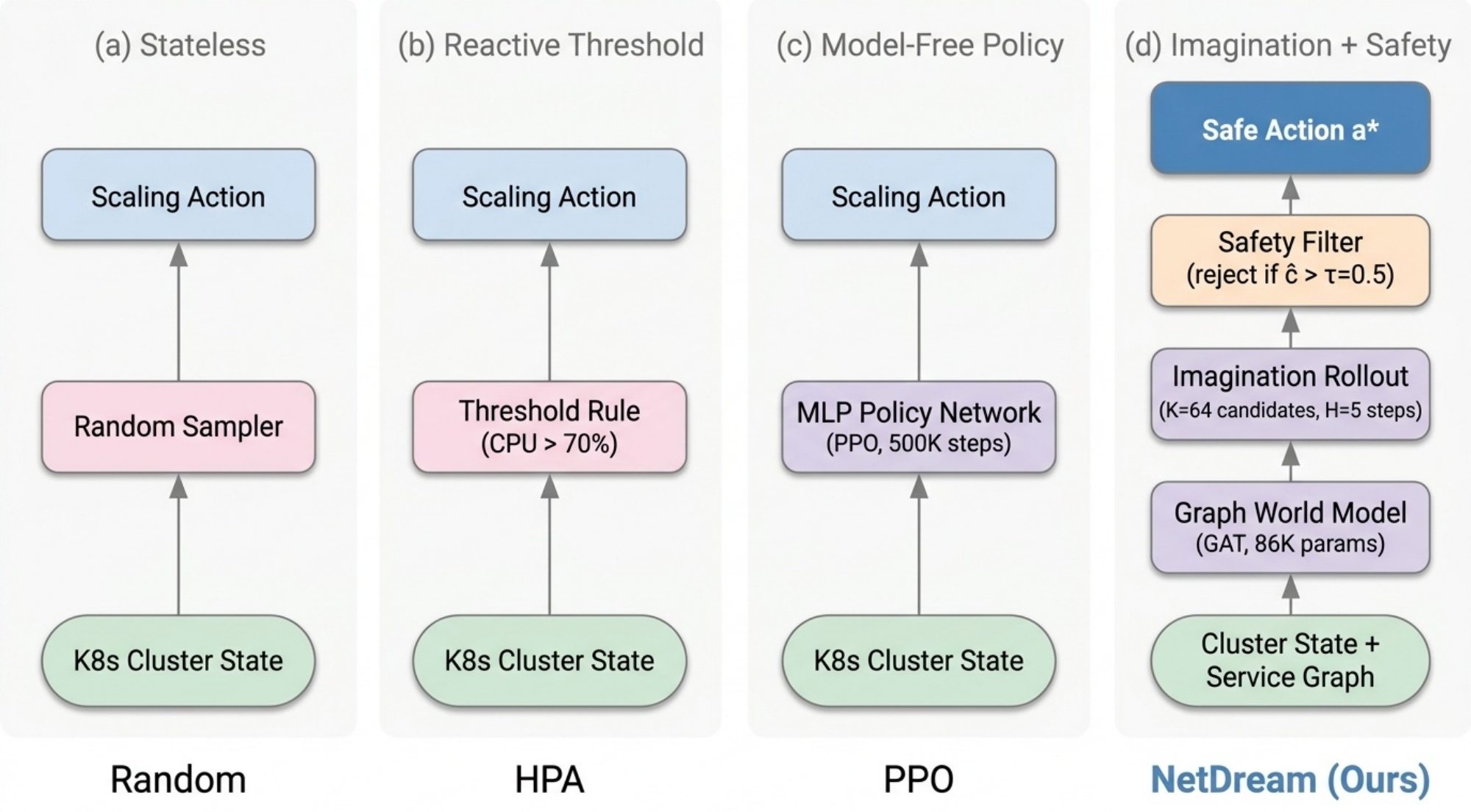
(a) Stateless
Scaling Action
Random Sampler
K8s Cluster State
Random
(b) Reactive Threshold
Scaling Action
Threshold Rule
(CPU > 70%)
K8s Cluster State
HPA
(c) Model-Free Policy
Scaling Action
MLP Policy Network
(PPO, 500K steps)
K8s Cluster State
PPO
(d) Imagination + Safety
Safe Action a*
Safety Filter
(reject if ĉ > τ=0.5)
Imagination Rollout
(K=64 candidates, H=5 steps)
Graph World Model
(GAT, 86K params)
Cluster State +
Service Graph
NetDream (Ours)